

Product Requirements Document (PRD)

Project: AI-Assisted Estimation & Pricing Engine

Company: OneSyntax **Version:** 1.5 (Updated with INVEST Framework & Controlled Transparency)

1. Project Overview

The primary objective of this system is to generate highly accurate, risk-managed pricing estimations for software development projects. This system acts as an "Estimation Assistant" rather than a fully autonomous AI. It utilizes AI technology strictly for scope breakdown (Task generation), hidden risk identification, and suggesting baseline 3-point estimates. Final approvals, calculations (PERT, RSD), and pricing decisions are governed by human intervention (Human-in-the-loop) and hard-coded deterministic logic.

2. Core Principles (Strict Guidelines)

- **No AI Math:** The AI (LLM) will not perform any mathematical calculations (Expected Time, SD, RSD). All calculations must be executed deterministically by the backend using hard-coded formulas.
- **Human-in-the-loop (Review & Approve):** No AI-generated output (WBS task list or 3-Point Estimates) is accepted as final. A Tech Lead must review, override/edit if necessary, and explicitly "Approve" all data before it proceeds to the next stage.
- **Controlled Transparency:** Raw Internal data (Optimistic, Most Likely, Pessimistic values, and RSD percentages) are strictly confidential and must never be exposed to the client. Client-facing transparency is maintained by sharing the detailed scope (WBS), a calculated pricing range, and clearly defined "Assumptions & Exclusions."

3. System Architecture & Modules

Module 1: AI Work Breakdown Structure (WBS) Generator & Human Approval

- **Input:** Client Requirement Document (PDF, Text) or Project Brief.
- **Process (AI LLM):** The AI breaks down the entire project into smaller technical tasks. The System Prompt must instruct the AI to strictly follow the Agile **INVEST framework**:
 - **I (Independent):** Tasks should be as independent as possible to avoid overlapping dependencies.
 - **N (Negotiable):** Tasks should capture the essence of what needs to be done, leaving room for implementation details.
 - **V (Valuable):** Each task must deliver value to the client or the system.
 - **E (Estimable):** Tasks must be clear enough to allow for accurate time estimation (Crucial for this engine).
 - **S (Small):** Tasks should be small enough to be completed within a few days or a single sprint.
 - **T (Testable):** The completion of the task must be verifiable.
- **Process (User Action - Tech Lead):**
 - The Tech Lead reviews the AI-generated Task List.
 - Adds missing tasks, deletes hallucinations, or edits existing tasks.

- Clicks "Approve" to finalize the WBS.
- **Output:** The Approved Task List.

Module 2: AI Risk Flagger

- **Process (AI LLM):** Analyzes every approved task from Module 1 to identify potential technical risks, edge cases, security vulnerabilities, and 3rd-party API limitations.
- **Output:** Generates specific "Risk Notes" for each task (This helps the Tech Lead accurately determine the Pessimistic/Worst-case scenario).

Module 3: AI 3-Point Estimation & Human Approval

- **Process (AI Generation):** Based on the approved WBS and Risk Notes, the AI suggests baseline Optimistic (O), Most Likely (M), and Pessimistic (P) values (in hours or days) using industry averages.
- **Process (User Action - Tech Lead):**
 - The Tech Lead reviews the AI-suggested estimates alongside the Risk Notes.
 - Overrides the AI's numbers based on team velocity, legacy code conditions, or specific expertise.
 - Clicks "Approve" to lock in the final O, M, and P values.

Module 4: Deterministic Math Engine (Calculation)

- **Process (Code/Backend):** Calculates the PERT and Risk metrics based on the approved values.
- **Formulas:**
 - **Expected Time (E)** = $(O + 4M + P) / 6$
 - **Standard Deviation (SD)** = $(P - O) / 6$
 - **Relative Standard Deviation (RSD)** = $(SD / E) * 100$

Module 5: Pricing Strategy Decision Engine

- **Process:** Recommends the safest pricing strategy based on the overall project RSD.
- **Logic:**
 - **RSD < 10%:** [GREEN FLAG] "Low Risk." Recommend a Fixed Price contract.
 - **RSD 10% - 20%:** [YELLOW FLAG] "Medium Risk." A Fixed Price can be offered, but a mandatory Risk Buffer (e.g., Expected Time + 1 SD) must be added to the final quote.
 - **RSD > 20%:** [RED FLAG] "High Risk!" Strictly refuse a Fixed Price contract. Recommend Time & Materials (T&M) or Hourly Rate agreements only.

Module 6: Client-Facing Proposal Generator (Safe Transparency)

- **Process:** Generates a professional, client-ready proposal that hides internal metrics while maintaining trust through scope clarity.
- **Output (What the client sees):**
 1. **Detailed Scope (WBS):** The comprehensive task list to show exactly what is being built.
 2. **Assumptions & Exclusions:** AI-identified technical risks translated into business conditions (e.g., "Assumes 3rd-party API documentation is 100% accurate. Delays due to API downtimes are excluded and will be billed separately").

3. Transparent Pricing Model:

- Displays a **Calculated Expected Range** instead of a single fixed number (e.g., \$5,000 - \$6,500).
- Includes a disclaimer: *"This estimate is subject to change based on the assumptions listed above."*

4. Technical Stack Considerations

- **Frontend:** React, Vue, or Svelte (Essential for dynamic, real-time recalculation of E, SD, and RSD as the Tech Lead adjusts sliders/inputs).
- **Backend:** Node.js / Python (To handle deterministic math and orchestrate AI API calls).
- **AI Integration:** OpenAI API (GPT-4o) or Anthropic Claude. **Crucial:** API calls must utilize "Structured Output (JSON mode)" to ensure seamless integration with the frontend UI.

5. Future Enhancements

- **Historical Data Analysis:** Feed actual completion times of finished projects back into the system to calculate developer/team estimation bias. This will allow the AI to calibrate future baseline estimates automatically.